Shaan Pakala

Website: https://shaanpakala.github.io/ | Email: shaan.pakala@gmail.com
Google Scholar, LinkedIn & GitHub | Phone: (408) 891-0158

About

I am a 1st year Computer Science Ph.D. student at the University of California, Riverside, where I work on AI4Science research problems with my advisor Professor Vagelis Papalexakis. This includes developing interpretable surrogate models for designing new materials, or efficiently generating expensive physics simulation data using generative machine learning.

Education

University of California, Riverside

- **Ph.D. in Computer Science** — **Since Sept. 2025**
- **B.S. in Data Science** — **Sept. 2021 – June 2025**

Research Experience

Lawrence Livermore National Laboratory

- **Graduate Research Intern** — **Summer 2026**
  Intern in the predictive healthcare group, with Dr. Priyadip Ray, working on machine learning methods for early disease outbreak detection
- **Graduate Research Intern** — **Summer 2025**
  Intern in the predictive healthcare group, with Dr. Priyadip Ray and Dr. Braden Soper, worked on causal inference for Parkinson's Disease progression

University of California, Riverside

- **Graduate Student Researcher** — **Since June 2025**
  Research with Prof. Papalexakis, working on AI4Science problems using tensor decomposition: (i) surrogate modeling for robust & interpretable material design [1] [3] and (ii) efficient physics simulation data generation using tensor decomposition in GANs & diffusion models [4]
- **Undergraduate Student Researcher** — **June 2024 – June 2025**
  Worked on automating the optimization of data science pipelines (hyperparameter optimization, neural network architecture search, SQL query cardinality estimation) with tensor completion [2]

Papers *(* denotes equal contribution)*

Conference

[1] **Tensor Methods: A Unified and Interpretable Approach for Material Design**
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
ACM SIGKDD (AI for Sciences track) 2026 [PDF] [Code]

[2] **Automating Data Science Pipelines with Tensor Completion**
Shaan Pakala, B. Graw, D. Ahn, T. Dinh, M.T. Mahin, V. Tsotras, J. Chen, and E.E. Papalexakis
IEEE International Conference on Big Data 2024 [Link] [PDF] [Code]

Workshop

[3] **Surrogate Modeling for the Design of Optimal Lattice Structures using Tensor Completion**
Shaan Pakala, A.E. Gongora, B. Giera, and E.E. Papalexakis
NeurIPS AI for Accelerated Material Design Workshop 2025 [PDF] [Code]

[4] **Efficiently Generating Multidimensional Calorimeter Data with Tensor Decomposition Parameterization**
P. Goulart*, Shaan Pakala*, and E.E. Papalexakis
ICCV Representation Learning with Very Limited Resources Workshop 2025 [PDF] [Code]

Awards

- **Travel Awards** (3): ACM KDD'26, NeurIPS'25 AI4Mat Workshop, and IEEE BigData'24
- **Undergraduate Research Spotlight Award** 2025
  Bourns College of Engineering – University of California, Riverside
- **NSF Research Experience for Undergraduates** 2024
  University of California, Riverside

Contributed Talks & Posters

Talks

- **Tensor Methods for Material Design** 2026
  Lawrence Livermore National Laboratory Annual CASIS Workshop
- **Surrogate Modeling with Tensor Completion** 2026
  UTRGV MECIS Spring Webinar

Posters

- **Tensor Methods for Interferometer Simulations** 2026
  CECI Workshop on Detector Science
- **Tensor Methods for Material Design** 2026
  SAIR Science & AI Summit
- **Tensor Completion for Material Property Prediction** 2025
  AAAI Bridge on Knowledge-guided Machine Learning

Academic Service

Organizer

- AI4AutoSci Workshop at IEEE BigData'26 2026

Reviewer

- TensorKDD Workshop at ACM KDD'26 2026
- AI Data Scientist Workshop at ACM KDD'26 2026